

BANKING & LOAN ANALYTICS

Project Overview

My project analyzes 100,000 loan application records to understand loan demand, approval performance, and repayment risk. I used Python, SQL, and Power BI to clean, analyze, and visualize the data. The main goal is to help the bank make better loan approval and risk-monitoring decisions.

Tools & Technologies

- **Programming:** Python
- **Data Analysis:** Pandas, NumPy
- **Data Visualization:** Matplotlib, Seaborn
- **Database & Analysis:** SQL
- **Business Intelligence:** Power BI, DAX
- **Data Sources:** Excel, CSV

Project Roadmap

- **Business Problem → Data Understanding → Data Cleaning → Feature Engineering → SQL Analysis → Python EDA → Power BI Dashboard → Business Insights & Recommendations.**

1. Dataset Summary

Rows: 100,000 | Original Columns: 14 | New Analytical Columns: 6

Key Features

- **Customer:** Age, Gender, City
- **Employment & Income:** Employment Type, Annual Income
- **Credit:** Credit Score
- **Loan:** Loan Type, Loan Amount, Loan Tenure
- **Application:** Application Date, Approval Status
- **Repayment:** Loan Status, Payment Delay Days
- **Derived:** Age Group, Income Segment, Credit Score Band, Delay Category, Application Year, Application Month

2 Data Understanding

Checked dataset structure, data types, missing values, duplicates, and basic statistics using Python functions like `shape()`, `info()`, `isnull()`, `duplicated()`, and `describe()`.

3 Data Quality Checking

Checked **missing values**, **duplicates**, **data types**, **invalid values**, and **consistency** to ensure the dataset was reliable for analysis.

4 Data Cleaning

The dataset was cleaned and validated by:

- Handling missing values and duplicates — `isnull()`, `dropna()`, `duplicated()`, `drop_duplicates()`
- Standardizing **City** and **Employment_Type** — `str.strip()`, `str.title()`
- Validating **Loan_Amount** and **Application_Date** — `pd.to_numeric()`, `pd.to_datetime()`
- Checking **Approval_Status** and **Loan_Status** — `value_counts()`, `unique()`
- Validating **Credit_Score**, **Payment_Delay_Days**, and data types — `describe()`,

5 Feature Engineering

Created 6 new analytical features to make the dataset easier to analyze:

- **Application_Year** — Extracted the year from **Application_Date**.
- **Application_Month** — Extracted the month from **Application_Date**.
- **Age_Group** — Categorized customers into different age groups.
- **Income_Segment** — Grouped customers based on **Annual_Income**.
- **Credit_Score_Band** — Categorized customers based on **Credit_Score**.
- **Delay_Category** — Categorized customers based on **Payment_Delay_Days**.

6. Python EDA Charts

Chart Name	Why We Create It
1. Loan Applications by Year	To understand how loan demand changes over time.
2. Loan Applications by Loan Type	To identify which loan types have the highest demand.
3. Approved vs Rejected	To understand the overall loan approval and rejection pattern.
4. Loan Amount Distribution	To understand the typical loan amount range customers apply for.
5. Credit Score Distribution	To understand the overall credit quality of applicants.
6. Loan Status Distribution	To understand the overall health of the loan portfolio, including overdue/default loans.
7. Payment Delay Distribution	To understand customer repayment behavior and identify delay patterns.
8. Average Loan Amount by Employment Type	To compare loan size across different employment groups.
9. Credit Score vs Loan Amount (Optional)	To check whether there is any relationship between credit score and requested loan amount.

7 Export Data to MySQL

Exported the cleaned dataset from Python/Pandas to MySQL using `to_sql()`, verified the 100,000 records, and prepared the data for SQL analysis.

8 SQL Analysis

1. Total number of loan applications?
2. Total loan amount?
3. Average loan amount?
4. Minimum loan amount?
5. Maximum loan amount?
6. Average credit score?
7. Applications for each loan type?
8. Total loan amount for each loan type?
9. Average loan amount for each loan type?
10. Which loan type has the highest applications?
11. Which loan type has the highest total loan amount?
12. Average loan tenure for each loan type?
13. How many applications were approved?
14. How many applications were rejected?
15. Overall approval rate?
16. Approval rate for each loan type?
17. Approval rate for each employment type?
18. Approval rate by gender?
19. Approval rate by credit score band?

20. Approval status by income segment?
21. Applications in each age group?
22. Applications in each income segment?
23. Which cities have the highest applications?
24. Average income for each employment type?
25. Average loan amount for each employment type?
26. Average credit score for each employment type?
27. Distribution of loan status?
28. Overdue and default loans for each loan type?
29. Average payment delay for each loan type?
30. Which credit score bands have the highest high-delay customers?

9 Power BI Dashboard Structure

Dashboard Page	Meaning	Main Purpose
Page 1 — Executive Overview	Overall bank loan performance	Understand total applications, total loan amount, approval performance, and loan demand.
Page 2 — Loan Approval Analysis	Who gets approved and why	Understand which loan types and customer segments have better approval performance.
Page 3 — Risk & Repayment Analysis	Where the bank has repayment risk	Identify overdue/default loans and customers with higher payment delays.
Page 4 — Customer & Loan Analysis	Who the bank's customers are	Understand customer age, income, credit score, employment, city, and loan patterns.

10 . KPI Plan

Total KPIs: 16 | 4 KPIs per dashboard page

Page 1 — Executive Overview

KPI	Meaning	Formula	Business Purpose
KPI 1: Total Applications	Bank ko total kitne loan applications receive hue.	COUNT(Customer_ID)	Overall loan demand measure karne ke liye.

KPI 2: Total Loan Amount	All loan applications ka total requested loan amount.	SUM(Loan_Amount)	Bank ke total loan demand/exposure ko measure karne ke liye.
KPI 3: Approval Rate	Total applications mein se kitne applications approve hue.	(Approved Applications / Total Applications) × 100	Bank ki overall approval performance measure karne ke liye.
KPI 4: Average Loan Amount	Ek application ka average loan amount.	AVERAGE(Loan_Amount)	Customers generally kitne amount ka loan apply kar rahe hain, ye samajhne ke liye.

Page 2 — Loan Approval Analysis

KPI	Meaning	Formula	Business Purpose
KPI 5: Approved Applications	Kitne loan applications approve hue.	COUNT where Approval_Status = "Approved"	Total successful loan applications measure karne ke liye.
KPI 6: Rejected Applications	Kitne loan applications reject hue.	COUNT where Approval_Status = "Rejected"	Loan rejection volume understand karne ke liye.
KPI 7: Approval Rate	Total applications mein approved applications ka percentage.	(Approved Applications / Total Applications) × 100	Bank ki loan approval efficiency compare karne ke liye.
KPI 8: Average Credit Score	All applicants ka average credit score.	AVERAGE(Credit_Score)	Applicant customer base ki overall credit quality samajhne ke liye.

Page 3 — Risk & Repayment Analysis

KPI	Meaning	Formula	Business Purpose
KPI 9: Overdue Loans	Kitne loans overdue status mein hain.	COUNT where Loan_Status = "Overdue"	Bank ke current repayment risk ko measure karne ke liye.
KPI 10: Default Loans	Kitne loans default status mein hain.	COUNT where Loan_Status = "Default"	Serious repayment risk identify karne ke liye.
KPI 11: Average Payment Delay	Customers average kitne days payment delay kar rahe hain.	AVERAGE(Payment_Delay_Days)	Overall repayment behavior aur delay level understand karne ke liye.
KPI 12: High Delay Customers	60 days se zyada payment delay karne wale customers.	COUNT where Payment_Delay_Days > 60	High-risk repayment customers identify karne ke liye.

Page 4 — Customer & Loan Analysis

KPI	Meaning	Formula	Business Purpose
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KPI 13: Average Age	Loan applicants ki average age.	AVERAGE(Age)	Bank ke typical customer profile ko understand karne ke liye.
KPI 14: Average Income	Applicants ki average annual income.	AVERAGE(Annual_Income)	Customer financial profile aur earning capacity understand karne ke liye.
KPI 15: Average Credit Score	Applicants ka average credit score.	AVERAGE(Credit_Score)	Customer base ki overall credit quality understand karne ke liye.
KPI 16: Average Loan Tenure	Customers average kitne years ke loan tenure ke liye apply kar rahe hain.	AVERAGE(Loan_Tenure_Years)	Customers ke typical loan repayment period ko understand karne ke liye.

11 . Power BI Dashboard Charts

Page 1 — Executive Overview

Chart Name	Why We Create It
1. Applications by Year	To understand the yearly loan demand trend.
2. Applications by Loan Type	To identify the most demanded loan types.
3. Approved vs Rejected	To understand overall approval performance.
4. Loan Amount by Loan Type	To identify which loan types generate the highest loan amount.

Page 2 — Loan Approval Analysis

Chart Name	Why We Create It
5. Approval Rate by Loan Type	To identify which loan types have better approval performance.
6. Approval Status by Employment Type	To compare approval and rejection patterns across employment groups.
7. Approval Status by Gender	To compare approval and rejection patterns across gender groups.
8. Approval Rate by Credit Score Band	To understand how approval performance varies with credit quality.

Page 3 — Risk & Repayment Analysis

Chart Name	Why We Create It
9. Loan Status Distribution	To understand the overall repayment status of loans.
10. Default/Overdue by Loan Type	To identify loan types with higher repayment risk.

11. Average Payment Delay by Loan Type	To identify which loan types have higher payment delays.
12. Risk by Credit Score Band	To identify which credit-score groups have higher default/overdue risk.

Page 4 — Customer & Loan Analysis

Chart Name	Why We Create It
13. Applications by Age Group	To identify the main customer age groups.
14. Applications by Income Segment	To understand which income segments contribute most applications.
15. Loan Amount by Employment Type	To compare total loan demand across employment groups.
16. Applications by City	To identify cities with higher loan demand.

6. Dashboard Meaning — One-Line Summary

Dashboard	Business Question
Dashboard 1 — Executive Overview	How is the bank's overall loan business performing?
Dashboard 2 — Loan Approval Analysis	Which customers and loan types have better approval performance?
Dashboard 3 — Risk & Repayment Analysis	Where is the bank facing higher repayment risk?
Dashboard 4 — Customer & Loan Analysis	Who are the bank's main customers and what loan patterns do they show?

Banking-Loan-Analytics



